

Comments on

Fully accounted NLO electromagnetic radiative corrections to the width of J/ψ decay

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Equation and table numbers refer to the current draft (`w_jpsi-4`). “VPK note” refers to *Radiative Corrections to the J/ψ photoproduction reaction*.

1 Eq. (1) and Eq. (17): replace E_V by m_V

Both read $1/(2E_V)$. The whole calculation is done in the J/ψ rest frame, where $E_V = m_V$ identically, and the text already says so. Writing m_V removes a symbol that never takes any other value, and makes the normalisation of Eq. (1) directly comparable with the standard two-body form.

2 Eq. (6): use the factorised form, and add the ultrarelativistic limit

Eq. (6) is correct — it is algebraically identical to Eq. (7) of the VPK note (verified symbolically; the difference is exactly zero). But the factorised form is easier to read and to check dimensionally, because the mass dependence enters only through dimensionless ratios:

$$\Gamma_B = \frac{4\pi\alpha^2}{f_V^2} \frac{m_V}{3} \left(1 + \frac{2m_\ell^2}{m_V^2}\right) \sqrt{1 - \frac{4m_\ell^2}{m_V^2}}.$$

Please also add the ultrarelativistic limit, Eq. (8) of the VPK note, which is the form actually used to fix f_V from the measured leptonic width:

$$\Gamma_B = \frac{4\pi\alpha^2}{f_V^2} \frac{m_V}{3}.$$

Worth stating explicitly: the correction connecting the two is *second* order in the mass ratio, not first. With $x = m_\ell^2/m_V^2$,

$$(1 + 2x)\sqrt{1 - 4x} = 1 - 6x^2 + O(x^3),$$

so the linear term cancels. This is why the massless limit is so accurate: 4×10^{-15} for $J/\psi \rightarrow e^+e^-$ and 8×10^{-6} for $J/\psi \rightarrow \mu^+\mu^-$. It reaches only 2×10^{-3} for $\rho \rightarrow \mu^+\mu^-$.

3 Table I: please say what $m_V = 1.5$ GeV is

Tables I and II are computed at $m_V = 1.5$ GeV, which is neither the J/ψ nor any physical vector meson. A reader meeting it just after Section II, which sets $m_V = 3.0969$ GeV, will be confused.

We understand it to be the last row of Table I of Ref. [10], whose tabulated column is $2E$ (MeV), running from 500 to 1500 with ρ , ω and ϕ marked — that is, $2E$ is the meson mass, since $E = m_V/2$ at resonance (stated in [10] below their Eq. (22)). One sentence would settle it: “for comparison with Ref. [10] we take $m_V = 1.5$ GeV, the largest mass tabulated there.”

4 The comparison with Ref. [10] does not hold where it is claimed

The text states that Table I can be “successfully compared” with Table I of Ref. [10] at $\omega/m_e = 0.01$, last line of the first half. At that line the two disagree, and the disagreement grows with r_{cut} :

r_{max}	Ref. [10]	this work	difference
0.10	−0.087	−0.0866	~ 0
0.25	−0.034	−0.0330	0.001
0.50	−0.002	+0.000044	0.002
1.00	+0.021	+0.023891	0.0029

The soft part agrees exactly (both give $\delta_R = -0.747$), so the mass and the cut-off are matched correctly. The discrepancy is entirely in the hard part and equals $5\alpha/4\pi = 0.0029$ at $r_{max} = 1$.

The reason is that **Eq. (24) of Ref. [10] is missing a term**. Their differential spectrum, Eq. (23), is correct and agrees with Eq. (14) of this work; the term was lost when that spectrum was integrated analytically. Their Eq. (26), and their Table I, inherit the omission. Recomputing their table from their own Eq. (26) reproduces every printed digit, electrons and muons alike, which confirms the diagnosis.

Suggested replacement for the claim of agreement:

Our Eq. (14) agrees with expression (23) of [10]. The integrated form, Eq. (24) of [10], is however missing a term $(\alpha/\pi)(3r_{max}/2 - r_{max}^2/4)$, so that the totals tabulated in Table I of [10] are low by that amount; at $r_{max} = 1$ the deficit is $5\alpha/4\pi = 0.0029$. Our results agree with [10] in the small- r_{max} rows, where the missing term lies below the quoted precision.

5 Please add the closed-form total correction, for both channels

The paper gives δ_{self} (8), δ_{vert} (12), δ_{soft} (13), and δ_{hard} only as an unevaluated integral (14), then the exact result (28). There is no compact closed form for the total correction, which is what an experiment actually applies and what a reader comes to this paper looking for. We would like it in the paper, written out separately for the two channels. These are Eqs. (19) and (20) of the VPK note, i.e. Eq. (26) of Ref. [10] with the missing term restored.

Electrons, $J/\psi \rightarrow e^+e^-$:

$$\delta(m_V, k_{max}) = -\frac{4\alpha}{\pi} \left[\left(\frac{1}{2} - \ln \frac{m_V}{m_e} \right) \left(\ln r_{max} + \frac{1}{3} \right) + \frac{1}{2} \left(L_2(r_{max}) - \frac{\pi^2}{6} \right) + \frac{17}{72} + R(r_{max}) + \frac{r_{max}^2}{16} - \frac{3}{8} r_{max} \right]$$

Muons, $J/\psi \rightarrow \mu^+\mu^-$:

$$\delta(m_V, k_{max}) = -\frac{4\alpha}{\pi} \left[\left(\frac{1}{2} - \ln \frac{m_V}{m_\mu} \right) \left(\ln r_{max} + \frac{1}{3} \right) + \frac{1}{2} \left(L_2(r_{max}) - \frac{\pi^2}{6} \right) + \frac{17}{72} \right. \\ \left. + \left(\frac{5}{18} - \frac{1}{3} \ln \frac{m_V}{m_e} \right) + R(r_{max}) + \frac{r_{max}^2}{16} - \frac{3}{8} r_{max} \right]$$

with, in each case,

$$r_{max} = \frac{2k_{max}}{m_V}, \quad R(r_{max}) = \left(r_{max} - \frac{r_{max}^2}{4} - \frac{3}{4} \right) \left[\ln \left(\frac{m_V}{m_\ell} \sqrt{1 - r_{max}} \right) - \frac{1}{2} \right], \quad L_2(x) = - \int_0^x \frac{\ln(1-t)}{t} dt,$$

m_ℓ being the mass of the lepton in the channel concerned. Here r_{max} is Ehlutzky's variable, defined below his Eq. (25) as “the maximum fraction of the lepton energy E ... which is lost ... by the emission of a hard photon”, i.e. $r_{max} = k_{max}/E$ with $E = m_V/2$. It is identical to the $r_{cut} = \omega_{cut}/E_1$ already defined in this paper, so no new variable is needed — but see Section 6. The extra term in the muon expression, $\frac{5}{18} - \frac{1}{3} \ln \frac{m_V}{m_e}$, is the electron-loop vacuum polarization; it is exactly $-\frac{\pi}{4\alpha}$ times δ_{self} of Eq. (8) evaluated with m_e , and it correctly keeps m_e , not m_μ .

The last two terms of each bracket, $r_{max}^2/16 - \frac{3}{8}r_{max}$, are the ones absent from Ref. [10].

Where it belongs

Section IV, immediately after Eq. (14) — the closed form *is* the evaluation of that integral, combined with Eqs. (8), (12) and (13). Three reasons it sits naturally there and nowhere else:

- Section IV opens by promising “compact expressions for all contributions to decay width: both for virtual and real bremsstrahlung ones.” Compact expressions are then given for every individual contribution, but not for the sum, which is the only combination that is physical.
- Eq. (16), $\delta_{FSIC} = 3\alpha/4\pi$, becomes an immediate corollary: it is the $r_{max} \rightarrow 1$ limit of the closed form, less δ_{self} .
- Figure 4 plots exactly this quantity for both channels. At present the reader is shown the curve without being given the function.

One caveat on placement. Section IV states that the URA is “applicable for $\ell = e$ ”, and that the mass-dependent treatment is “crucial for ... the muon channel”. That is consistent with the numbers below: the electron formula is exact for all practical purposes, the muon one is good to $\sim 10^{-4}$. We would therefore present the electron expression as the working formula of Section IV, and the muon expression alongside it with an explicit note that it carries the $(m_\mu/E)^2$ error quantified below — pointing the reader to Eq. (28) when that matters.

Status of the approximation

These expressions are *not* exact in the lepton mass, and they are also not the $m_\ell = 0$ limit. They are the ultrarelativistic result: the lepton mass is kept in the logarithms, while terms suppressed by $(m_\ell/E)^2$, $E = m_V/2$, are dropped. Ref. [10] states this explicitly just above its Eq. (22) (“...neglecting terms of order $(m/E)^2$ ”), and the same restriction propagates to its Eqs. (24) and (26). We suggest saying so when the formulas are introduced, since this paper's whole point is the mass-exact treatment.

The size of what is dropped:

	$(m_\ell/E)^2$	δ_{self} : Eq. (8) – Eq. (9)	total at $r_{max} = 1$: this form – Eq. (28)
e^+e^-	1.1×10^{-7}	$+3 \times 10^{-10}$	$+1 \times 10^{-7}$
$\mu^+\mu^-$	4.7×10^{-3}	$+1.1 \times 10^{-5}$	-7×10^{-5}

So for the electron channel the closed form is exact at any precision an experiment will reach, and for the muon channel it is good to $\sim 10^{-4}$ absolute — which is precisely the accuracy argument made in Section IV of the draft. Presenting the closed form alongside Eq. (28) would let a reader use the simple expression where it suffices and the exact one where it does not.

Checks the expressions pass:

- they reproduce Table I of this work to all six digits at every r_{cut} , in both channels, and are independent of ω by construction;
- at $r_{max} = 1$ they give exactly $\delta_{self} + 3\alpha/4\pi$, i.e. δ_{self} plus the FSIC of Eq. (16);
- for $J/\psi \rightarrow e^+e^-$ at $r_{max} = 1$ the value is 0.026135, matching the Bardin–Shumeiko result of Table III.

6 The upper limit of r_{cut} is not 1

The text before Eq. (16) says the cut energy takes its maximum value “at (15) or $r_{cut} = r_{max} = 1$ ”, and Table III is captioned ($r_{max} = 1$). That is inconsistent with Eq. (15) itself. With $r_{cut} = \omega_{cut}/E_1 = 2\omega_{cut}/m_V$ and $\omega_{max} = m_V/2 - 2m_\ell^2/m_V$,

$$r_{cut}^{\max} = \frac{2\omega_{max}}{m_V} = 1 - \frac{4m_\ell^2}{m_V^2} = \beta^2,$$

which is 0.99535 for muons, not 1. (For electrons it is $1 - 1.1 \times 10^{-7}$.)

This is not only notational. Eq. (29) defines $\rho_1(\omega_{cut}) = \sqrt{m_V(m_V - 2\omega_{cut})} + \sqrt{m_V(m_V - 2\omega_{cut}) - 4m_\ell^2}$. Setting $\omega_{cut} = m_V/2$, i.e. literally $r_{cut} = 1$, makes the second radicand $-4m_\ell^2 < 0$ and ρ_1 imaginary. At the true endpoint $\omega_{cut} = \omega_{max}$ that radicand vanishes exactly and $\rho_1(\omega_{max}) = 2m_\ell$, which is the correct and finite limit. So Eq. (28) is well defined up to ω_{max} and no further.

We take it that Table III was in fact evaluated at $\omega_{cut} = \omega_{max}$, which is the sensible thing to do; only the label is wrong. We suggest replacing “ $r_{cut} = r_{max} = 1$ ” and the Table III caption by “ $\omega_{cut} = \omega_{max}$ ” or “ $r_{cut} = \beta^2$ ”. The same remark applies to the upper end of the axis in Fig. 4, which is correctly drawn as $\omega_{cut}/\omega_{max}$.

7 Reference [9]

[9] P. Chatagnon, e-Print: 2602.22128 [hep-ex] (2026).

is now published. Please replace with

[9] P. Chatagnon, Phys. Rev. C 113 (2026) 6, 065203.

Additional point, for consideration

Eq. (4). As printed,

$$\sum_{\text{pol}} \varepsilon_{\alpha}(p_V) \varepsilon_{\beta}(p_V) = -g_{\alpha\beta}$$

is the polarization sum for a *massless* vector. For a massive vector meson it is

$$\sum_{\lambda} \varepsilon_{\alpha} \varepsilon_{\beta}^* = -g_{\alpha\beta} + \frac{p_{\alpha} p_{\beta}}{m_V^2}.$$

The extra term does drop here, because it contracts the lepton current with $p_V = p_1 + p_2$ and $\bar{u}(p_1)(\not{p}_1 + \not{p}_2)v(p_2) = 0$ by the Dirac equations — so Eq. (5) and everything after it are correct. But the identity as written is not true on its own, and the surrounding text calls it “the photon” while the sum is over the three J/ψ polarizations. A clause such as “the $p_{\alpha} p_{\beta}$ term is dropped by current conservation” would fix it.